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Studierichting: Informatica
Oefeningenreeks 7B nr 2.4

$$w = yz + zx + xy$$

$$x = s^2 - t^2 \Rightarrow \frac{\partial w}{\partial x} = z + y$$

$$y = s^2 + t^2 \Rightarrow \frac{\partial w}{\partial y} = z + x$$

$$z = s^2t^2 \Rightarrow \frac{\partial w}{\partial z} = y + x$$

$$\frac{\partial x}{\partial s} = 2s$$

$$\frac{\partial y}{\partial s} = 2s$$

$$\frac{\partial z}{\partial s} = 2st^2$$

$$\frac{\partial x}{\partial t} = -2t$$

$$\frac{\partial y}{\partial t} = 2t$$

$$\frac{\partial z}{\partial t} = 2ts^2$$

$$\frac{\partial w}{\partial s} = (z + y)(2s) + (z + x)(2s) + (y + x)(2st^2)$$

$$= (s^2t^2 + s^2 + t^2)(2s) + (s^2t^2 + s^2 - t^2)(2s) + (2s^2)(2st^2)$$

$$= 2s^3t^2 + 2s^3 + 2st^2 + 2s^3t^2 + 2s^3 - 2st^2 + 4s^3t^2$$

$$= 8s^3t^2 + 4s^3$$

$$\frac{\partial w}{\partial t} = (s^2t^2 + s^2 + t^2)(-2t) + (s^2t^2 + s^2 - t^2)(2t) + (2s^2)(2st^2)$$

$$= -2s^2t^3 - 2s^2t - 2t^3 + 2s^2t^3 + 2s^2t - 2t^3 + 4s^4t$$

$$= -4t^3 + 4s^4t$$

References